

Sutan Gading Fadhillah Nasution

Software Engineer

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PROFESSIONAL SUMMARY

Software Engineer with 5 years of experience in architecting scalable modern applications, specializing in Full-Stack & Frontend Engineering, Agentic AI, and Infrastructure. Proficient in TypeScript, Python, Docker, SQL, Kubernetes, and CI/CD, bridging clean modular code with resilient, scalable infrastructure.

WORK EXPERIENCE

Software Engineer (Part-Time) · Tridorian, Singapore (Remote) Jun 2026 – Present
Observability, performance, and reliability across a 6-service AI platform for medical-legal document analysis.

- Built End-to-End Tracing: Set up OpenTelemetry across 4 Python services and GCP Cloud Trace, building shared library components for decorators, context managers, and Pub/Sub propagation.
- Sped up report generation by 74% (35 to 9 minutes) using parallel AI fan-out and cut event extraction time by 81% (21 to 4 minutes).
- Audited 18 exception paths and fixed 9 silent failures, making sure errors actually surface in traces instead of getting swallowed.
- Eliminated a billing race condition by moving counter increments to the database, preventing parallel workers from overwriting usage counts and causing underbilling.

Full Stack Engineer (Freelance) · Lazone.id, Jakarta (Remote) Mar 2026 – Jun 2026
End-to-end full-stack development, from frontend, backend, and API design to CI/CD deployments.

- Integrated Midtrans payment gateway with idempotent webhooks and DB transactions to eliminate race conditions during ticket activation.
- Implemented end-to-end payment telemetry across transaction initiation, webhook processing, and activation flows, making failed payment states traceable across the full request lifecycle.
- Delivered the full platform within at least one month, owning architecture, implementation, deployment, and iteration through changing client requirements.
- Containerized core services with Docker and custom Bash scripts for automated staging and production deployments.
- Configured an NPM Workspaces monorepo isolating Next.js frontend, NestJS API, and shared schemas into modular packages.

Software Engineer (Full-Time) · Heypico.ai, Singapore (Remote)

Aug 2025 – Mar 2026

End-to-end development of AI agents, GraphRAG architecture, mobile apps, and custom device orchestration protocols.

- Engineered a GraphRAG system using Neo4j and Graphhitti with long-term memory modules to improve AI context retrieval accuracy.
- Developed backend APIs and tools for AI agents to execute multi-step automations and Model Context Protocol (MCP) integrations.
- Created automated AI agents for financial reporting in Google Docs/Sheets and transactional food ordering workflows.
- Built a custom Termux-based device package and remote control layer that resolved cloud-to-device connectivity issues for companion hardware.
- Spearheaded the MVP launch across Web, Android, and iOS, handling the entire iOS release lifecycle through App Store review.
- Authored 10+ reusable AI middleware utilities, reducing integration setup time across engineering squads.

Software Engineer (Full-Time) · Avataralabs.ai, Jakarta (Remote)

Sep 2022 – Aug 2025

Frontend engineering and full-stack development across Generative AI platforms, web3 wallet infrastructure, and NFT ecosystems.

- Developed an AI Researcher Dashboard using RAG for personalization, driving 3x revenue growth and 5x client acquisition across livestreams and embedded chat tools.
- Led frontend architecture across 4 AI-driven products, including a content platform featuring TTS, STT, video generation, and real-time chat interfaces.
- Built end-to-end custodial and semi-custodial Web3 wallet solutions alongside a custom Web3 Provider SDK to streamline smart contract interactions for clients.
- Designed and implemented NFT landing pages, dashboards, and wallet UI flows in collaboration with backend and design teams.

Frontend Engineer (Full-Time) · Sirclo-Orami, Tangerang

Dec 2019 – Nov 2022

Frontend ecosystem engineering, design system development, and e-commerce core features across web and mobile platforms.

- Built key e-commerce features including an online store for kids' apparel, social commerce group buying, and Express Mart for local shop discovery.
- Partnered with Lead Frontend Engineers on R&D initiatives, architectural evaluations, and frontend utilities to support multiple development squads.
- Overhauled merchant Seller Dashboard features, improving product management workflows for active e-commerce sellers.

SKILLS

Backend	TypeScript, NodeJS, NestJS, Python, FastAPI, PostgreSQL, Redis
Web & Mobile	TypeScript, React, NextJS, Tailwind CSS, Tanstack Router & Query
Cloud, Infra & DevOps	Docker, Kubernetes, Proxmox, Cloudflare, GCP, AWS, CI/CD (Github Actions)
AI & Agentic Systems	LLM Integrations, MCP Server Creation, RAG & Vector Search

PROJECTS

Nouverse Platform · Creator · app.nouverse.tech

2026 – Present

A Cloud Application Platform (Vercel-like PaaS) that automates full-stack deployments, custom domain routing, SSL provisioning, and multi-tenant billing.

- Automated cloud hosting with instant builds, isolated container orchestration, and dynamic Traefik/Cloudflare ingress for automatic SSL provisioning.
- Built a multi-tenant payment gateway and subscription system with real-time usage tracking and webhooks handling.
- Designed a scalable server architecture configured to seamlessly join and scale within a Kubernetes cluster.

Nouride AI Harness · Creator · nouride.com

2026 – Present

An ultra-lightweight, self-hosted multi-agent AI runtime for personal assistants (like OpenClaw or Hermes) designed for edge devices and homelabs (down to a 512MB Raspberry Pi Zero 2). Features a single-binary footprint, deterministic tool output spilling to protect context limits, and native multi-platform messaging (Telegram, WhatsApp, Discord).

- Architected a lightweight, single-binary daemon (~60MB) with a 64MB RAM baseline footprint and 0.7% single-core idle CPU, enabling 24/7 multi-agent operation on sub-512MB RAM hardware.
- Implemented an integrated Human-in-the-Loop (HITL) approval gate and privilege ceiling to sandbox shell execution (`exec`) and prevent unauthorized system modifications.
- Implemented a deterministic Tool Output Spilling engine that auto-offloads oversized tool results to sandboxed disk storage, preserving context limits while allowing agents to paginate and inspect large outputs via read/grep.

Agent Memory Engine & MCP · Creator · gading.dev/blog/building-agent-memory-without-vendor-lock-in-real-world-implementation

2026 – Present

An agent-agnostic open-source Model Context Protocol (MCP) server providing a vendor-agnostic long-term memory engine, hybrid RAG search, and modular skills architecture across multiple AI agents and IDEs (OpenClaw, Cursor, Zed, Claude Code).

- Architected a vendor-agnostic long-term memory MCP server using Python, pgvector, and dual transport protocols (Stdio/SSE).
- Built a hybrid retrieval system pairing pgvector semantic search with SQL analytics over daily summaries and automated fallbacks.
- Implemented markdown export integrations for personal knowledge management systems and graph visualization tools.

LANGUAGES

Indonesian

Native

English

Professional working

LINKS

Blog

gading.dev/blog

GitHub

github.com/gadingnst

Linkedin

linkedin.com/in/gadingnst